



Caringbah High School

Year 12 2023

Mathematics Extension 2

HSC Course

Assessment Task 4 – Trial HSC Examination

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I 10 marks

Attempt Questions 1-10

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 90 marks

Attempt Questions 11-16

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II						Total
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16	
/10	/15	/15	/15	/15	/15	/15	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. What is the quadratic equation with solutions $2 + 3i$ and $2 - 3i$?

(A) $z^2 - 4z + 13 = 0$

(B) $z^2 - 4z - 13 = 0$

(C) $z^2 + 4z - 5 = 0$

(D) $z^2 + 4z - 13 = 0$

2. Which of the following is an expression for $\int \frac{x}{\sqrt[3]{x^2 + 1}} dx$?

(A) $\frac{3}{2} \sqrt[3]{(x^2 + 1)^2} + C$

(B) $\frac{1}{2} \sqrt[3]{(x^2 + 1)^2} + C$

(C) $\frac{3}{4} \sqrt[3]{(x^2 + 1)^2} + C$

(D) $\frac{3}{2} \sqrt[3]{(x^2 + 1)^2} + C$

3. What is the size of the acute angle θ between the vectors

$\vec{a} = 2\vec{i} - \vec{j} - \vec{k}$ and $\vec{b} = 2\vec{i} - 2\vec{k}$?

(A) $\theta = \frac{\pi}{6}$

(B) $\theta = \frac{\pi}{5}$

(C) $\theta = \frac{\pi}{3}$

(D) $\theta = \frac{\pi}{4}$

4. Which of the following statements is TRUE?

- (A) $\forall a, b \in \mathbb{R} \quad \sin a < \sin b \Rightarrow a < b$
(B) $\forall a, b \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \sin a < \sin b \Rightarrow a < b$
(C) $\forall a, b \in \mathbb{R} \quad \cos a < \cos b \Rightarrow a < b$
(D) $\forall a, b \in [0, \pi] \quad \cos a < \cos b \Rightarrow a < b$

5. Each pair of lines given below intersects at $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

Which pair of lines are perpendicular?

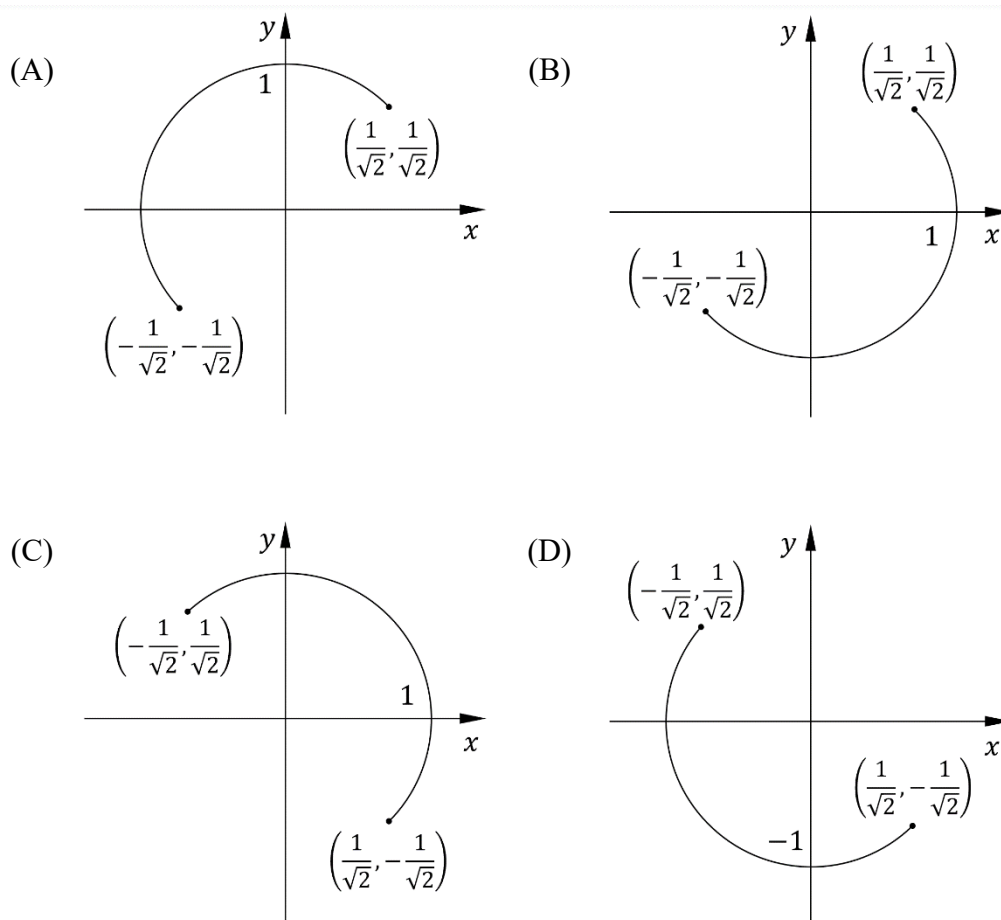
- (A) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$
(B) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$
(C) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$
(D) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 3 \\ -2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \\ -5 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$

6. A particle moves in simple harmonic motion such that $v^2 + 9x^2 = k$.
What is the period of the particle's motion ?

- (A) $\frac{2\pi}{k}$
(B) 3π
(C) $\frac{3k}{2\pi}$
(D) $\frac{2\pi}{3}$

7. Which diagram best shows the curve described by the position vector

$$\tilde{r}(t) = \sin(t) \tilde{i} - \cos(t) \tilde{j} \text{ for } \frac{\pi}{4} \leq t \leq \frac{5\pi}{4} ?$$



8. Which expression is equal to

$$\int \left(x^2 \frac{d}{dx} (x^3 - 1) + (x^3 - 1) \frac{d}{dx} (x^2) \right) dx ?$$

- (A) $\frac{x(x^2 - 1)}{6} + C$
- (B) $\frac{x^3}{3} \times \frac{x^4 - x}{4} + C$
- (C) $x^2(x^3 - 1) + C$
- (D) $2x(3x^2 - 1) + C$

9. What are the values of real numbers p and q such that $(2 - i)$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = -11$ and $q = -20$
- (B) $p = -11$ and $q = 20$
- (C) $p = 11$ and $q = -20$
- (D) $p = 11$ and $q = 20$
10. The non-zero complex numbers ω and z are linked by the formula $\omega = z + k\bar{z}$, where k is real and $k \neq 0$.
- Which of the following statements is INCORRECT?
- (A) ω can only be purely imaginary if $k = -1$
- (B) ω can only be purely real if $k = 1$
- (C) ω and z must have different moduli
- (D) ω and z must have different arguments

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) If $z = 4 - i$, express the following in the form $a + ib$ where a and b are real.

(i) $\overline{iz}$ 1

(ii) $\frac{1}{z}$ 1

(b) Find the value of $\int_0^{\sqrt{5}} \frac{x}{\sqrt{x^2 + 4}} dx$. 2

(c) The point A has position vector $\overrightarrow{OA} = 2\hat{i} + 6\hat{j} - 3\hat{k}$ relative to an origin O . 2
Find a unit vector parallel to $\overrightarrow{OA}$.

(d) (i) Find the values of a , b and c given

$$\frac{1}{x(x^2 + 1)} \equiv \frac{a}{x} + \frac{bx + c}{x^2 + 1}. \quad 2$$

(ii) Hence, or otherwise, evaluate

$$\int_1^2 \frac{dx}{x(x^2 + 1)} \quad 2$$

(e) (i) Find the two square roots of $2i$, giving the answers in the form $x + iy$, 2
where x and y are real numbers.

(ii) Hence, or otherwise, solve $2z^2 + 2\sqrt{2}z + 1 - i = 0$, leaving answers in 3
the form $x + iy$.

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the equation $z^5 + 1 = 0$. Find the roots of this equation and show them on an Argand diagram. 3

- (b) For all non-negative numbers, x and y , $\frac{x+y}{2} \geq \sqrt{xy}$. (Do NOT prove this.) 4

A rectangle has dimensions a and b .

Given that the rectangle has perimeter P , and area A , prove that $P^2 \geq 16A$.

- (c) Use integration by parts to evaluate 3

$$\int_1^2 x e^{2x} dx.$$

- (d) A particle is moving in simple harmonic motion with its acceleration given by:

$$\ddot{x} = -12 \sin 2t$$

Initially, the particle is at the origin and has a positive velocity of 6 m/s.

- (i) Show the equation for the particle's velocity is $\dot{x} = -12 \sin^2 t + 6$. 2

- (ii) Show that $\ddot{x} = -4x$. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) By completing the square, find $\int \frac{1}{9x^2 + 6x + 5} dx$ **3**

(b) (i) Show that for any integer n , $e^{in\theta} - e^{-in\theta} = 2i \sin(n\theta)$. **2**

(ii) By expanding $(e^{i\theta} - e^{-i\theta})^3$, show that **3**

$$\sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$$

(c) Lines l_1 and l_2 are given below, relative to a fixed origin O .

$$l_1: \quad \tilde{r} = \left(11\tilde{i} + 2\tilde{j} + 17\tilde{k} \right) + \lambda \left(-2\tilde{i} + \tilde{j} - 4\tilde{k} \right)$$

$$l_2: \quad \tilde{r} = \left(-5\tilde{i} + 11\tilde{j} + p\tilde{k} \right) + \mu \left(-3\tilde{i} + 2\tilde{j} + 2\tilde{k} \right)$$

(i) Given that line l_1 and l_2 intersect, find the value of p . **3**

(ii) Hence find the point of intersection of line l_1 and l_2 . **1**

(d) Plot the following points on the Argand Plane and show that $z = 2i$, $w = \sqrt{3} - i$ and $v = -\sqrt{3} - i$ are vertices of an equilateral triangle. **3**

End of Question 13

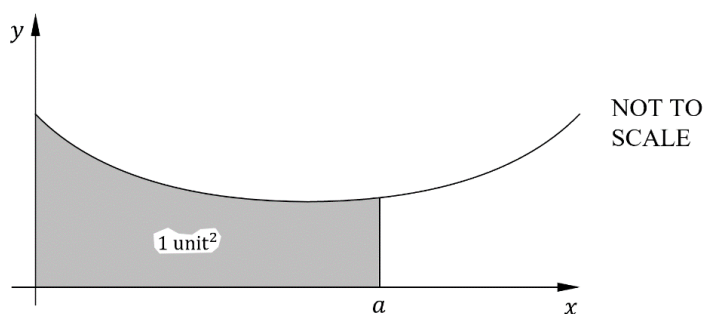
Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that the three cube roots of i are $z_1 = e^{\frac{\pi}{6}i}$, $z_2 = e^{\frac{5\pi}{6}i}$ and $z_3 = -i$, prove that: 4

$$(z_3)^2 = (z_1 i)^2 + (z_2 i)^2$$

- (b) Prove that $\log_2 n$ is irrational when n is an odd integer greater than or equal to 3. 3

- (c) The area under the curve $y = \frac{1}{1+\sin x}$ from 0 to $x = a$ is 1 unit². 4



Find the value of a .

- (d) Show that $\frac{x}{e} > \ln x$ for $x > e$. 2

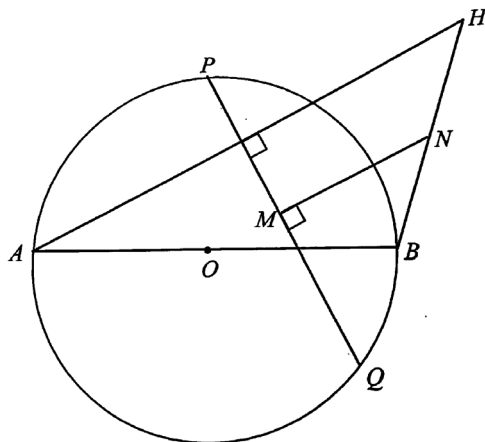
- (e) A particle is moving in simple harmonic motion along the x -axis with an amplitude of 3 metres. At time t seconds it has displacement x metres from the origin O and velocity v ms⁻¹ given by $v^2 = n^2(a^2 - (x - c)^2)$. 2

If the particle has a speed $2\sqrt{5}$ ms⁻¹ at the origin and a speed 6 ms⁻¹ when it is 2 metres to the right of the origin, what is its centre of motion?

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a)



In the diagram, AB is the diameter of a circle with centre O and PQ is a chord of the circle that is not perpendicular to AB . The perpendicular from A to PQ is produced to the point H outside the circle. M is the midpoint of PQ and N is the point on BH such that $MN \perp PQ$ and the points O, M and N are collinear.

Let $\vec{OA} = \underline{a}$, $\vec{OP} = \underline{p}$, $\vec{OQ} = \underline{q}$ and $\vec{AH} = \underline{h}$.

- (i) If $\vec{BN} = \lambda \vec{BH}$ for some scalar λ , show $\vec{ON}$ is $(2\lambda - 1)\underline{a} + \lambda\underline{h}$. 1
- (ii) Hence show that N is the midpoint of BH . 3

- (b) Let $z = x + iy$ and hence solve the equation $z^2 = |z|^2 - 4$ 3

- (c) Position vectors of the points A, B and C , relative to an origin O , are $-\underline{i} - \underline{j}$, $\underline{j} + 2\underline{k}$, and $4\underline{i} + \underline{k}$ respectively.

- (i) Find $\vec{AB}$. 1
- (ii) Find $|\vec{AB}|$. 1
- (iii) Prove that $\angle ABC$ is a right angle. 3

- (d) A particle of mass m is moving in a straight line under the action of a force. 3

$$F = \frac{m}{x^3}(6 - 10x)$$

What is the velocity in any position if the particle starts from rest at $x = 1$?

End of Question 15

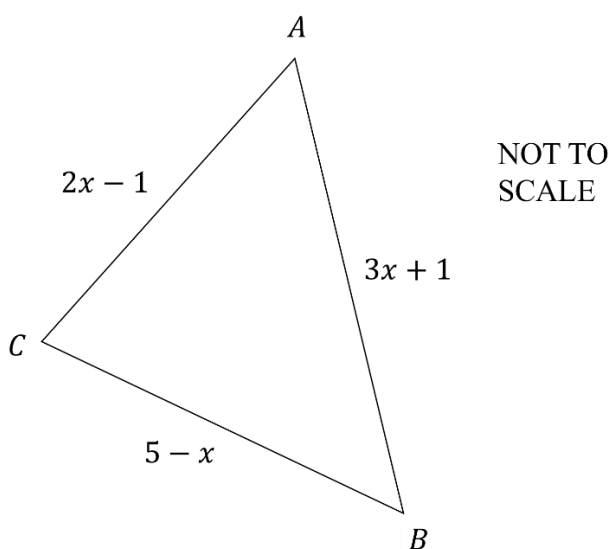
Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Let $I_n = \int_0^1 (1 - x^r)^n dx$ where $r > 0$ for $n = 1, 2, 3, \dots$

Show that $I_n = \frac{nr}{nr + 1} I_{n-1}$ 2

(ii) Hence or otherwise, find the exact value of $\int_0^1 (1 - x^{\frac{3}{2}})^3 dx$ 2

(b) Triangle ABC has sides $a = 5 - x$, $b = 2x - 1$ and $c = 3x + 1$ as shown, where $x \in \mathbb{R}$.



(i) Prove that $\frac{5}{6} < x < \frac{3}{2}$. 2

(ii) Given $\triangle ABC$ is isosceles, prove that $\cos A = \frac{1}{8}$. 2

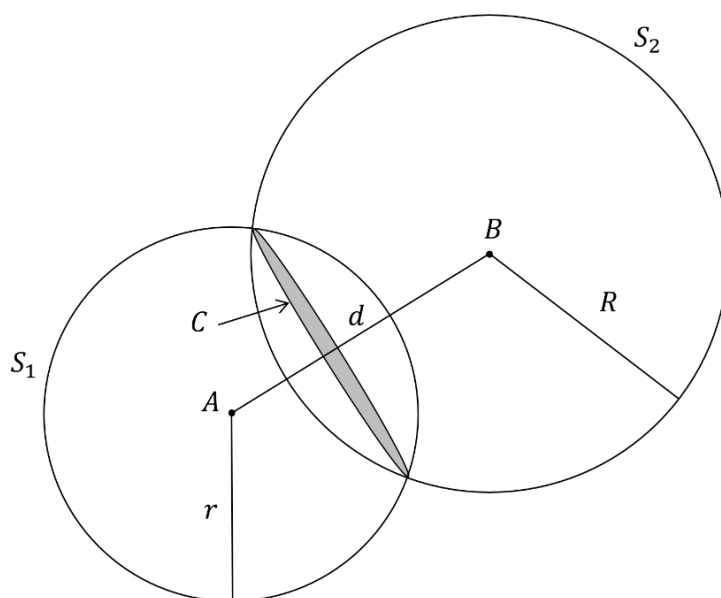
(c) Sketch the locus of z on the Argand diagram of the below and mark where the inequalities $|z - 1| \leq 3$ and $\text{Im}(z) \geq 3$ hold simultaneously. 2

Question 16 continues next page

- (d) Two spheres, S_1 and S_2 , intersect in the circle C .

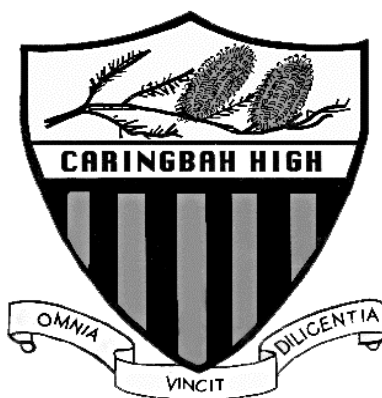
The sphere S_1 is centred at $A(a, b, c)$ with radius r and the sphere S_2 is centred at $B(a + m, b + n, c + p)$ with radius R , where a, b, c, m, n, p, r and R are real, with $r, R > 0$.

The distance between the centres of the spheres is d .



- (i) Prove that the circle C lies on the sphere S_3 , which has AB as a diameter, only if $m^2 + n^2 + p^2 = r^2 + R^2$. 4
- (ii) Hence, or otherwise, show that the intersection of the spheres $x^2 + y^2 + z^2 = 25$ and $(x - 3)^2 + (y - 4)^2 + (z - 12)^2 = 144$ is a circle lying on a third sphere, which has the interval joining the centres of the first two spheres as a diameter. 1

End of Examination



Caringbah High School

Year 12 2023

Mathematics Extension 2

HSC Course

Assessment Task 4 – Trial HSC Examination Solutions

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	C
3	A
4	B
5	A
6	D
7	C
8	C
9	B
10	D

Worked Solutions

1 A

$$(z - (2 + 3i))(z - (2 - 3i)) = 0$$

$$z^2 - 2z + 3iz - 2z - 3iz + 4 + 9 = 0$$

$$\therefore z^2 - 4z + 13 = 0, \text{ so A.}$$

2 C

$$\text{Let } u = x^2 + 1$$

$$\frac{du}{dx} = 2x \text{ and } xdx = \frac{1}{2} du$$

$$\begin{aligned} \int \frac{x}{\sqrt[3]{x^2 + 1}} dx &= \frac{1}{2} \int \frac{1}{\sqrt[3]{u}} du \\ &= \frac{1}{2} \int u^{-\frac{1}{3}} du \\ &= \frac{1}{2} \times \frac{3}{2} u^{\frac{2}{3}} + C \\ &= \frac{3}{4} \sqrt[3]{(x^2 + 1)^2} + C \end{aligned}$$

, so C.

3 A

$$\cos \theta = \frac{\tilde{a} \cdot \tilde{b}}{|\tilde{a}| |\tilde{b}|} = \frac{4 + 0 + 2}{\sqrt{6} \sqrt{8}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2} \quad \therefore \theta = \frac{\pi}{6}, \text{ so A.}$$

4 B

The correct answer must be an increasing function in the whole domain, which only occurs for B.

5 A

Looking for the dot product of the direction vectors being zero.

$$\text{A: } \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} = -2 + 2 + 0 = 0, \text{ so A.}$$

6 D

$$v^2 + 9x^2 = k$$

$$v^2 = k - 9x^2$$

$$\begin{aligned}\ddot{x} &= \frac{d}{dx} \left(\frac{v^2}{2} \right) \\ &= \frac{d}{dx} \left(\frac{k}{2} - \frac{9x^2}{2} \right) \\ &= -9x = -n^2x\end{aligned}$$

Hence $n = 3$

Period: $T = \frac{2\pi}{n} = \frac{2\pi}{3}$, so D.

7 C

$$r\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} \times i - \cos \frac{\pi}{4} \times j = \frac{1}{\sqrt{2}}i - \frac{1}{\sqrt{2}}j = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

$$r\left(\frac{5\pi}{4}\right) = \sin \frac{5\pi}{4} \times i - \cos \frac{5\pi}{4} \times j = -\frac{1}{\sqrt{2}}i - \left(-\frac{1}{\sqrt{2}}\right)j = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

Also passes through:

$$r\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} \times i - \cos \frac{\pi}{2} \times j = i = (1, 0), \quad \text{so C.}$$

8 C

The integrand involves the product rule

$$\int \left(x^2 \frac{d}{dx} (x^3 - 1) + (x^3 - 1) \frac{d}{dx} (x^2) \right) dx$$

$$= \int \frac{d}{dx} (x^2(x^3 - 1)) dx$$

$$= x^2(x^3 - 1) + c, \quad \text{so C.}$$

Alternatively:

$$\int \left(x^2 \frac{d}{dx} (x^3 - 1) + (x^3 - 1) \frac{d}{dx} (x^2) \right) dx$$

$$= \int (x^2 \times 3x^2 + (x^3 - 1) \times 2x) dx$$

$$= \int (3x^4 + 2x^4 - 2x) dx = \int (5x^4 - 2x) dx$$

$$= x^5 - x^2 + c$$

$$= x^2(x^3 - 1) + c, \quad \text{so C.}$$

9 B

Using the conjugate root theorem $(2 + i)$ and $(2 - i)$ are both roots of the equation $z^3 + pz + q = 0$.

$(2 + i) + (2 - i) + \alpha = 0$ (sum of the roots)

$$\alpha = -4$$

$(2 + i) \times (2 - i) \times (-4) = -q$ (product of the roots)

$$(4 + 1) \times (-4) = -q$$

$$q = 20$$

$(2 + i)(2 - i) + (2 - i) \times (-4) + (2 + i) \times (-4) = p$

$\therefore p = -11$ and $q = 20$, so B.

10 D

A: when $k = -1$, $z - 1(\bar{z}) = 2\text{Im}(z)$ which is purely imaginary

B: when $k = 1$, $z + 1(\bar{z}) = 2\text{Re}(z)$ which is purely real.

C: the modulus of z is $\sqrt{x^2 + y^2}$ while the modulus of ω is $\sqrt{(k + 1)^2 x^2 + (k - 1)^2 y^2}$, and the two moduli could only be equal if $k + 1 = 1 \rightarrow k = 0$ and $k - 1 = 1 \rightarrow k = 2$, which is a contradiction.

D: when z is purely real then its argument is 0 or π , and its conjugate, $\bar{z}$, is equal so has the same argument.

When added together the argument is unchanged, so ω has the same argument and so D is INCORRECT.

Section II

Question 11 (a) (i)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\begin{aligned}\bar{iz} &= \overline{i(4 - i)} = \overline{4i + 1} \\ &= 1 - 4i \quad \boxed{A}\end{aligned}$$

Question 11 (a) (ii)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\begin{aligned}\frac{1}{z} &= \frac{1}{4 - i} \times \frac{4 + i}{4 + i} = \frac{4 + i}{16 + 1} \\ &= \frac{4}{17} + \frac{1}{17}i \quad \text{or} \quad \frac{4 + i}{17} \quad \boxed{A}\end{aligned}$$

Question 11 (b)

Criteria	Marks
Provides correct solution	2
Sets up the integral in terms of u.	1

Sample answer:

Let $u = x^2 + 4$

$$\frac{du}{dx} = 2x \text{ or } \frac{1}{2} du = x dx$$

When $x = 0$ then $u = 4$ and when $x = \sqrt{5}$ then $u = 9$

$$\begin{aligned} \int_0^{\sqrt{5}} \frac{x}{\sqrt{x^2 + 4}} dx &= \frac{1}{2} \int_4^9 \frac{1}{\sqrt{u}} du = \frac{1}{2} \int_4^9 u^{-\frac{1}{2}} du \quad [A] \\ &= \frac{1}{2} \left[2u^{\frac{1}{2}} \right]_4^9 \\ &= [\sqrt{9} - \sqrt{4}] \\ &= 1 \quad [B] \end{aligned}$$

Question 11 (c)

Criteria	Marks
Provides correct solution	2
Finds the magnitude of $\overrightarrow{OA}$.	1

Sample answer:

$$\begin{aligned} |\overrightarrow{OA}| &= \sqrt{2^2 + 6^2 + (-3)^2} \\ &= \sqrt{49} = 7 \quad [A] \end{aligned}$$

$$\begin{aligned} \widehat{OA} &= \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|} \\ &= \frac{1}{7} (2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}) \quad [B] \end{aligned}$$

Question 11 (d) (i)

Criteria	Marks
Provides correct solution	2
Rearranges the numerator as $(a + b)x^2 + cx + a$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{1}{x(x^2 + 1)} &\equiv \frac{a}{x} + \frac{bx + c}{x^2 + 1} \\
 &\equiv \frac{a(x^2 + 1) + x(bx + c)}{x(x^2 + 1)} \\
 &\equiv \frac{(a + b)x^2 + cx + a}{x(x^2 + 1)} \quad \boxed{A}
 \end{aligned}$$

Equating coefficients $a + b = 0$, $c = 0$, $a = 1$ and thus $b = -1$

$$a = 1, b = -1, c = 0 \quad \boxed{B}$$

Question 11 (d) (ii)

Criteria	Marks
Provides correct solution	2
Find the correct primitive, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \int_1^2 \frac{dx}{x(x^2 + 1)} &= \int_1^2 \left(\frac{1}{x} - \frac{x}{x^2 + 1} \right) dx \\
 &= \int_1^2 \left(\frac{1}{x} - \frac{1}{2} \times \frac{2x}{x^2 + 1} \right) dx \\
 &= \left[\ln |x| - \frac{1}{2} \ln |x^2 + 1| \right]_1^2 \quad \boxed{A} \\
 &= \left(\ln 2 - \frac{1}{2} \ln 5 \right) - \left(\ln 1 - \frac{1}{2} \ln 2 \right) \\
 &= \frac{3}{2} \ln 2 - \frac{1}{2} \ln 5 \left(\text{or } \ln \left(\frac{2\sqrt{2}}{\sqrt{5}} \right) \right) \quad \boxed{B}
 \end{aligned}$$

Question 11 (e) (i)

Criteria	Marks
Provides correct solution	2
Finds one square root, or finds square roots in another form, or equivalent merit	1

Sample answer:

$$(a + ib)^2 = 2i$$

$$a^2 - b^2 + 2abi = 2i$$

$$a^2 - b^2 = 0 \quad \text{and} \quad ab = 1$$

$$a = \pm 1, b = \pm 1 \text{ by inspection} \quad \boxed{A}$$

$\therefore$ the two roots are $\pm(1 + i)$ $\boxed{B}$.

Question 11 (e) (ii)

Criteria	Marks
Provides correct solution	3
Simplifies the root of the discriminant to $\sqrt{8i}$ or equivalent merit	2
Correctly applies the quadratic formula, or equivalent merit	1

Sample answer:

$$z = \frac{-2\sqrt{2} \pm \sqrt{(2\sqrt{2})^2 - 4(2)(1-i)}}{2(2)} \quad \boxed{A}$$

$$= \frac{-2\sqrt{2} \pm \sqrt{8 - 8 + 8i}}{4}$$

$$= \frac{-2\sqrt{2} \pm 2\sqrt{2i}}{4} \quad \boxed{B}$$

$$= \frac{-\sqrt{2} \pm \sqrt{2i}}{2}$$

$$= \frac{-\sqrt{2} \pm (1 + i)}{2}$$

$$= -\frac{\sqrt{2} + 1}{2} - \frac{i}{2}, \quad \frac{1 - \sqrt{2}}{2} + \frac{i}{2} \quad \boxed{C}$$

Question 12 (a)

Criteria	Marks
Provides correct solution	3
Correctly finds all the roots, or equivalent merit	2
Correctly finds one root, or equivalent merit	1

Sample answer:

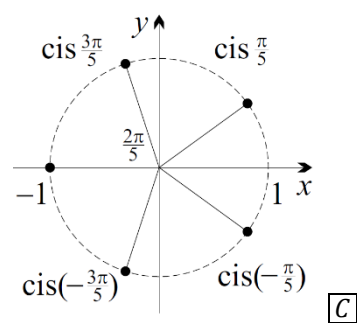
$$z^5 + 1 = 0 \quad \therefore z^5 = -1$$

$$\text{Let } z = \text{cis}\theta \quad \therefore z^5 = \text{cis}5\theta = -1$$

$$\therefore 5\theta = \pi \quad \therefore \theta = \frac{\pi}{5}$$

$$\therefore z_1 = \text{cis}\frac{\pi}{5} \quad \boxed{A}$$

$$\therefore z_2 = \text{cis}\frac{3\pi}{5}, z_3 = -1, z_4 = \text{cis}\frac{-\pi}{5}, z_5 = \text{cis}\frac{-3\pi}{5} \quad \boxed{B}$$



Question 12 (b)

Criteria	Marks
Provides correct solution	4
Correctly replaces A and P into the AM-GM	3
Substitutes a and b into the AM-GM, or equivalent merit	2
Finds expressions for the perimeter and area in terms of a and b, or equivalent merit	1

Sample answer:

$$P = 2(a + b), A = ab \quad \boxed{A}$$

$$\frac{a + b}{2} \geq \sqrt{ab} \quad \boxed{B}$$

$$\frac{2(a + b)}{4} \geq \sqrt{ab}$$

$$\frac{P}{4} \geq \sqrt{A} \quad \boxed{C}$$

$$\frac{P^2}{16} \geq A$$

$$P^2 \geq 16A \quad \boxed{D}$$

Question 12 (c)

Criteria	Marks
Provides correct solution	3
Finds the integral equals $\frac{2e^4}{2} - \frac{e^2}{2} - \frac{1}{2} \left[\frac{1}{2} e^{2x} \right]_1^2$	2
Differentiates one function and integrates the other	1

Sample answer:

$$\int_1^2 x e^{2x} dx \quad \boxed{A} \quad \begin{array}{ll} u = x & \frac{dv}{dx} = e^{2x} \\ \frac{du}{dx} = 1 & v = \frac{1}{2} e^{2x} \end{array}$$

$$= \left[\frac{x e^{2x}}{2} \right]_1^2 - \frac{1}{2} \int_1^2 e^{2x} dx \quad \boxed{B}$$

$$= \frac{2e^4}{2} - \frac{e^2}{2} - \frac{1}{2} \left[\frac{1}{2} e^{2x} \right]_1^2$$

$$= e^4 - \frac{e^2}{2} - \frac{e^4 - e^2}{4} \quad \boxed{C}$$

$$= \frac{3e^4}{4} - \frac{e^2}{4}$$

$$= \frac{e^2(3e^2 - 1)}{4}$$

Question 12 (d) (i)

Criteria	Marks
Provides correct solution	2
Finds velocity but does not clear constant, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \ddot{x} &= -12 \sin 2t \\
 \dot{x} &= -12 \int \sin 2t \, dt \\
 &= -12 \int 2 \sin t \cos t \, dt \\
 &= -12 \sin^2 t + C \quad [A]
 \end{aligned}$$

When $t = 0$, $\dot{x} = 6$ hence $C = 6$
 $\therefore \dot{x} = -12 \sin^2 t + 6 \quad [B]$

OR

$$\begin{aligned}
 \dot{x} &= -12 \int \sin 2t \, dt \\
 &= -12 \left[-\frac{\cos 2t}{2} \right] + C \\
 &= 6 \cos 2t + C \quad [A]
 \end{aligned}$$

When $t = 0$, $\dot{x} = 6$ hence $C = 0$
 $\therefore \dot{x} = 6 \cos 2t = 6(1 - 2 \sin^2 t)$
 $= -12 \sin^2 t + 6 \quad [B]$

Question 12 (d) (ii)

Criteria	Marks
Provides correct solution	3
Makes significant progress towards the solution.	2
Finds $x = 3 \sin 2t + C$ or equivalent merit.	1

Sample answer:

$$\begin{aligned}
 x &= \int -12 \sin^2 t + 6 \, dt \\
 &= \int -12 \left(\frac{1 - \cos 2t}{2} \right) + 6 \, dt = \int -6 + 6 \cos 2t + 6 \, dt \\
 &= \int 6 \cos 2t \, dt \\
 &= 3 \sin 2t + C \quad [A]
 \end{aligned}$$

When $t = 0$, $x = 0$ hence $C = 0$
 $\therefore x = 3 \sin 2t \quad [B]$

Then
 $\ddot{x} = -12 \sin 2t$
 $= -4 \times 3 \sin 2t$
 $= -4x \quad [C]$

Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Makes significant progress towards the solution.	2
• Completes the square or equivalent merit.	1

Sample answer:

$$\begin{aligned}
 \int \frac{1}{9x^2 + 6x + 5} dx &= \int \frac{dx}{9 \left[\left(x^2 + \frac{2}{3}x \right) \right] + 5} \\
 &= \int \frac{dx}{9 \left[\left(x + \frac{1}{3} \right)^2 - \frac{1}{9} \right] + 5} \quad [A] \\
 &= \int \frac{dx}{9 \left(x + \frac{1}{3} \right)^2 + 4} \\
 &= \int \frac{dx}{(3x + 1)^2 + 2^2} \quad [B] \\
 &= \frac{1}{3} \int \frac{3dx}{(3x + 1)^2 + 2^2} \\
 &= \frac{1}{6} \tan^{-1} \frac{(3x + 1)}{2} + C \quad [C]
 \end{aligned}$$

Question 13 (b) (i)

Criteria	Marks
Provides correct solution	2
Rewrites as the difference of conjugates or converts into mod-arg form	1

Sample answer:

$$e^{in\theta} - e^{-in\theta} = e^{in\theta} - \overline{e^{in\theta}} \quad [A]$$

$$= 2\text{Im}(e^{in\theta})$$

$$= 2i \sin n\theta \quad [B]$$

Alternative solution:

$$e^{in\theta} - e^{-in\theta} = (\cos n\theta + i \sin n\theta) - (\cos(-n\theta) + i \sin(-n\theta)) \quad [A]$$

$$= (\cos n\theta + i \sin n\theta) - (\cos(n\theta) - i \sin(n\theta))$$

$$= 2i \sin n\theta \quad [B]$$

Question 13 (b) (ii)

Criteria	Marks
Provides correct solution	3
Finds both equations marked as (1) and (2), or equivalent merit	2
Correctly uses the binomial expansion (without simplifying) or uses the result in (i) to show $(e^{i\theta} - e^{-i\theta})^3 = (2i \sin \theta)^3$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 (e^{i\theta} - e^{-i\theta})^3 &= (e^{i\theta})^3 + 3(e^{i\theta})^2(-e^{-i\theta}) + 3(e^{i\theta})(-e^{-i\theta})^2 + (-e^{-i\theta})^3 \quad [A] \\
 &= e^{i3\theta} - 3e^{-i\theta} + 3e^{-i\theta} - e^{-i3\theta} \\
 &= (e^{i3\theta} - e^{-i3\theta}) - 3(e^{i\theta} - e^{-i\theta}) \\
 &= 2i \sin 3\theta - 6i \sin \theta \quad (1)
 \end{aligned}$$

From (i):

$$\begin{aligned}
 (e^{i\theta} - e^{-i\theta})^3 &= (2i \sin \theta)^3 \\
 &= 8i^3 \sin^3 \theta \\
 &= -8 \sin^3 \theta \quad (2) \quad [B] \text{ or } [A] \text{ (without (1))}
 \end{aligned}$$

From (1) and (2):

$$-8 \sin^3 \theta = 2 \sin 3\theta - 6 \sin \theta$$

$$\sin^3 \theta = \frac{3 \sin \theta + \sin 3\theta}{4} \quad [C]$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Finds the correct values for λ and μ , or equivalent merit	2
• Correctly uses the binomial expansion (without simplifying) or uses the result in (i) to show $(e^{i\theta} - e^{-i\theta})^3 = (2 \sin \theta)^3$, or equivalent merit	1

Sample answer:

Line l_1 intersects the line l_2 then:

$$\begin{aligned} (11\hat{i} + 2\hat{j} + 17\hat{k}) + \lambda(-2\hat{i} + \hat{j} - 4\hat{k}) \\ = (-5\hat{i} + 11\hat{j} + p\hat{k}) + \mu(-3\hat{i} + 2\hat{j} + 2\hat{k}) \quad [A] \end{aligned}$$

$$11 - 2\lambda = -5 - 3\mu \quad (1)$$

$$2 + \lambda = 11 + 2\mu \quad (2)$$

$$17 - 4\lambda = p + 2\mu \quad (3)$$

Equation (1) + 2 $\times$ (2) $15 = 17 + \mu$ then $\mu = -2$

Equation (2) $2 + \lambda = 11 - 4$ then $\lambda = 5$ [B]

$$\begin{aligned} \text{Equation (3)} \quad 17 - 4 \times 5 &= p + 2 \times -2 \\ p &= 17 - 20 + 4 \\ &= 1 \end{aligned}$$

$\therefore$ The value of p is 1. [C]

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

From 13(b)(i) $\lambda = 5$, $\mu = -2$ and $p = 1$

$l_1: (11\hat{i} + 2\hat{j} + 17\hat{k}) + 5(-2\hat{i} + \hat{j} - 4\hat{k}) = \hat{i} + 7\hat{j} - 3\hat{k}$ or

$$l_2: (-5\hat{i} + 11\hat{j} + \hat{k}) - 2(-3\hat{i} + 2\hat{j} + 2\hat{k}) = \hat{i} + 7\hat{j} - 3\hat{k}$$

$\therefore$ Point of intersection is $\hat{i} + 7\hat{j} - 3\hat{k}$ [A]

Question 13 (d)

Criteria	Marks
Provides correct solution	3
Finds the length of zw or zv or wv, or equivalent merit	2
Plots the points or equivalent merit	1

Sample answer:

Pythagoras theorem

$$zw = \sqrt{3^2 + (\sqrt{3})^2}$$

$$= \sqrt{12} = 2\sqrt{3}$$

$$zv = \sqrt{3^2 + (\sqrt{3})^2}$$

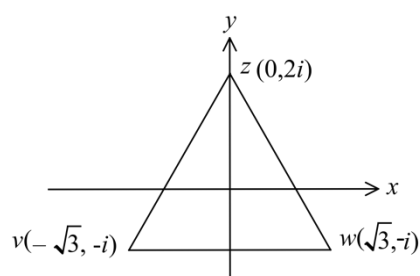
$$= \sqrt{12} = 2\sqrt{3}$$

$$wv = \sqrt{3} + \sqrt{3}$$

$$= 2\sqrt{3}$$

$\therefore$ Equilateral triangle.

[B]



[A]

[C]

Question 14 (a)

Criteria	Marks
Provides correct solution	4
Proves $(z_1 i)^2 + (z_2 i)^2 = -1$, or $(z_1)^2 + (z_2)^2 + (z_3)^2 = 0$, or equivalent merit	3
Uses conjugate rules or mod-arg form to simplify the sum of two complex numbers, or equivalent merit	2
Uses de Moivres Theorem, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 (z_1 i)^2 + (z_2 i)^2 &= i^2 \left(\left(e^{\frac{\pi i}{6}} \right)^2 + \left(e^{\frac{5\pi i}{6}} \right)^2 \right) \\
 &= - \left(e^{\frac{\pi i}{3}} + e^{\frac{5\pi i}{3}} \right) && \boxed{A} \\
 &= -e^{i\pi} \left(e^{-\frac{2\pi i}{3}} + e^{\frac{2\pi i}{3}} \right) \\
 &= -(-1) \left(e^{\frac{2\pi i}{3}} + e^{\frac{2\pi i}{3}} \right) \\
 &= 1 \times 2 \operatorname{Re} \left(e^{\frac{2\pi i}{3}} \right) && \boxed{B} \\
 &= 2 \cos \left(\frac{2\pi}{3} \right) \\
 &= 2 \left(-\frac{1}{2} \right) \\
 &= -1 && \boxed{C} \\
 &= (-i)^2 \\
 &= (z_3)^2 && \boxed{D}
 \end{aligned}$$

Question 14 (b)

Criteria	Marks
• Provides correct solution	3
• Rearranges to find $(2k + 1)^q = 2^p$, or equivalent merit	2
• Defines $\log_2(2k + 1) = \frac{p}{q}$, or equivalent merit	1

Sample answer:

Suppose $\log_2 n$ is rational.

Let $n = 2k + 1$ for $k \in \mathbb{Z}^+$

$$\therefore \log_2(2k + 1) = \frac{p}{q} \text{ for } p, q \in \mathbb{Z}, HCF(p, q) = 1, q \neq 0 \quad \boxed{A}$$

$$\therefore q \log_2(2k + 1) = p$$

$$\log_2(2k + 1)^q = p$$

$$(2k + 1)^q = 2^p \quad \boxed{B}$$

LHS is odd, since an odd number to any power is odd *.

RHS is even, since an even number to any power is even *.

We have a contradiction, so $\log_2 n$ must be irrational for all odd n . $\boxed{C}$

* Must mention that an odd number to any power is odd, not just LHS is odd, and similarly for the RHS being even.

Question 14 (c)

Criteria	Marks
Provides correct solution	4
Finds the value of the integral as $2\left(1 - \frac{1}{1+\tan\frac{a}{2}}\right)$ or equivalent	3
Simplifies the integrand to $(1+t)^{-2}$	2
Correctly substitutes for t including changing the limits and replacing dx	1

Sample answer:

$$\int_0^a \frac{dx}{1 + \sin x} \qquad t = \tan \frac{x}{2} \rightarrow dx = \frac{2dt}{1+t^2}$$

$$= \int_0^{\tan \frac{a}{2}} \frac{1}{1 + \frac{2t}{1+t^2}} \times \frac{2dt}{1+t^2} \quad [A]$$

$$= 2 \int_0^{\tan \frac{a}{2}} \frac{dt}{1+t^2+2t}$$

$$= 2 \int_0^{\tan \frac{a}{2}} (1+t)^{-2} dt \quad [B]$$

$$= 2 \left[\frac{(1+t)^{-1}}{-1} \right]_0^{\tan \frac{a}{2}} = 2 \left[\frac{1}{1+t} \right]_{\tan \frac{a}{2}}^0$$

$$= 2 \left(1 - \frac{1}{1 + \tan \frac{a}{2}} \right) \quad [C]$$

$$\therefore 2 \left(1 - \frac{1}{1 + \tan \frac{a}{2}} \right) = 1$$

$$1 - \frac{1}{1 + \tan \frac{a}{2}} = \frac{1}{2}$$

$$\frac{1}{1 + \tan \frac{a}{2}} = \frac{1}{2}$$

$$1 + \tan \frac{a}{2} = 2$$

$$\tan \frac{a}{2} = 1$$

$$\frac{a}{2} = \frac{\pi}{4}$$

$$\therefore a = \frac{\pi}{2} \quad [D]$$

Question 14 (d)

Criteria	Marks
Provides correct solution	2
Uses $f(e) = 0$ to deduce required inequality.	1

Sample answer:

$$\text{Let } f(x) = \frac{x}{e} - \ln x \text{ for } x > e$$

Hence $f(e) = 0$ and $f(x)$ is an increasing function $x > e$ A

$$\therefore f(x) > 0 \text{ for } x > e$$

$$\therefore \frac{x}{e} > \ln x \text{ for } x > e \quad \text{ B }$$

Question 14 (e)

Criteria	Marks
Provides correct solution	2
Write a simultaneous equation, or equivalent merit	1

Sample answer:

$$x = 0, \quad v^2 = 20 \quad \therefore n^2(9 - c^2) = 20 \quad - \textcircled{1}$$

$$x = 2, \quad v^2 = 36 \quad \therefore n^2(9 - (2 - c)^2) = 36 \quad - \textcircled{2} \quad \text{ A }$$

$$\therefore \frac{\textcircled{1}}{\textcircled{2}} = \frac{(9 - c^2)}{(9 - (2 - c)^2)} = \frac{20}{36}$$

$$\therefore \frac{(9 - c^2)}{(5 + 4c - c^2)} = \frac{5}{9}$$

$$\therefore c^2 + 5c - 14 = 0$$

$$\therefore (c - 2)(c + 7) = 0, \quad \therefore c = 2 \text{ (as } c > 0) \quad \text{ B }$$

Question 15 (a) (i)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\overrightarrow{ON} = \overrightarrow{OB} + \overrightarrow{BN} = -\underline{a} + \lambda \overrightarrow{BH} = -\underline{a} + \lambda (\overrightarrow{BA} + \overrightarrow{AH}) = -\underline{a} + \lambda (2\underline{a} + \underline{h})$$

$$\therefore \overrightarrow{ON} = -\underline{a} + \lambda (2\underline{a} + \underline{h}) = (2\lambda - 1)\underline{a} + \lambda \underline{h} \quad \boxed{A}$$

Question 15 (a) (ii)

Criteria	Marks
Provides correct solution.	3
Substantive progress ie. correct procedure but fails to explain $2\lambda - 1 = 0$.	2
Some progress ie. uses the perpendicularity to write an appropriate dot product.	1

Sample answer:

$$MN \perp PQ \quad \therefore \underline{h} \cdot (\underline{q} - \underline{p}) = 0 \quad \boxed{A}$$

and O, M and N are collinear $\therefore \overrightarrow{ON} \cdot \overrightarrow{PQ} = 0$ since $MN \perp PQ$

$$\therefore ((2\lambda - 1)\underline{a} + \lambda \underline{h}) \cdot (\underline{q} - \underline{p}) = 0$$

$$\therefore (2\lambda - 1)\underline{a} \cdot (\underline{q} - \underline{p}) = 0$$

But AB, PQ are not perpendicular. $\boxed{B}$

$$\therefore \underline{a} \cdot (\underline{q} - \underline{p}) \neq 0$$

$$\text{Hence } (2\lambda - 1) = 0 \quad \therefore \lambda = \frac{1}{2}$$

$$\therefore \overrightarrow{BN} = \frac{1}{2} \overrightarrow{BH} \text{ and hence N is the midpoint of BH} \quad \boxed{C}$$

Question 15 (b)

Criteria	Marks
Provides correct solution	3
Makes significant progress towards the solution.	2
Writes the equation using z^2 in terms of $x + iy$ or equivalent merit.	1

Sample answer:

Let $z = x + iy$

Then $z^2 = (x + iy)^2 = x^2 + 2ixy - y^2$ A

$$z^2 = |z|^2 - 4$$

$$x^2 + 2ixy - y^2 = |x + iy|^2 - 4$$

$$x^2 + 2ixy - y^2 = \left(\sqrt{x^2 + y^2}\right)^2 - 4$$

$$x^2 + 2ixy - y^2 = x^2 + y^2 - 4$$

$$2ixy - y^2 = y^2 - 4$$

$$2y^2 - 2ixy - 4 = 0$$
 B

$$2(y^2 - ixy - 2) = 0$$

$$(y^2 - 2) = 0 \text{ then } y = \pm\sqrt{2}$$

$$ixy = 0 \text{ then } x = 0$$

Therefore

$$z = x + iy$$

$$z = \pm\sqrt{2}i$$
 C

Question 15 (c) (i)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\begin{aligned}
 \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\
 &= (\underline{j} + 2\underline{k}) - (-\underline{i} - \underline{j}) \\
 &= \underline{i} + 2\underline{j} + 2\underline{k} \quad \boxed{A}
 \end{aligned}$$

Question 15 (c) (ii)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\begin{aligned}
 |\overrightarrow{AB}| &= \sqrt{x^2 + y^2 + z^2} \\
 &= \sqrt{1^2 + 2^2 + 2^2} \\
 &= 3 \quad \boxed{A}
 \end{aligned}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	3
• Uses the angle between two vectors.	2
• Finds $\overrightarrow{BC}$ or $ \overrightarrow{BC} $.	1

Sample answer:

$$\begin{aligned}
 \overrightarrow{BC} &= \overrightarrow{OC} - \overrightarrow{OB} \\
 &= (4\mathbf{i} + \mathbf{k}) - (\mathbf{j} + 2\mathbf{k}) \\
 &= 4\mathbf{i} - \mathbf{j} - \mathbf{k} \quad \boxed{A}
 \end{aligned}$$

$$\begin{aligned}
 |\overrightarrow{BC}| &= \sqrt{x^2 + y^2 + z^2} \\
 &= \sqrt{4^2 + (-1)^2 + (-1)^2} \\
 &= 3\sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 \cos \angle CAB &= \frac{\overrightarrow{AB} \cdot \overrightarrow{BC}}{|\overrightarrow{AB}| |\overrightarrow{BC}|} \quad \boxed{B} \\
 &= \frac{1 \times 4 + 2 \times (-1) + 2 \times (-1)}{3 \times 3\sqrt{2}} \\
 &= \frac{0}{9\sqrt{2}} = 0
 \end{aligned}$$

$$\angle CAB = 90^\circ \quad \boxed{C}$$

Question 15 (d)

Criteria	Marks
Provides correct solution	3
Integrates and uses the initial conditions to find an expression for $\frac{1}{2}v^2$.	2
Uses $v \frac{dv}{dx}$ for acceleration.	1

Sample answer:

$$\begin{aligned}
 F &= \frac{m}{x^3}(6 - 10x) \\
 ma &= \frac{m}{x^3}(6 - 10x) \\
 v \frac{dv}{dx} &= \frac{6}{x^3} - \frac{10}{x^2} \quad [A] \\
 \int v dv &= \int \left(\frac{6}{x^3} - \frac{10}{x^2} \right) dx \\
 \frac{1}{2}v^2 &= \left(\frac{6x^{-2}}{-2} - \frac{10x^{-1}}{-1} \right) + C \\
 \frac{1}{2}v^2 &= \left(\frac{-3}{x^2} + \frac{10}{x} \right) + C \\
 \text{When } v = 0 \text{ and } x = 1 \\
 \frac{1}{2}0^2 &= \left(\frac{-3}{1^2} + \frac{10}{1} \right) + C \\
 C &= -7 \quad [B] \\
 \text{Hence} \\
 \frac{1}{2}v^2 &= \left(\frac{-3}{x^2} + \frac{10}{x} \right) - 7 \\
 v^2 &= \left(\frac{-6}{x^2} + \frac{20}{x} \right) - 14 \\
 &= \frac{-6 + 20x - 14x^2}{x^2} \\
 v &= \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)} \quad [C]
 \end{aligned}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Sets up the integration by parts.	1

Sample answer:

$$\begin{aligned}
 I_n &= \int_0^1 (1 - x^r)^n dx \\
 &= [x(1 - x^r)^n]_0^1 - n \int_0^1 x(1 - x^r)^{n-1}(-rx^{r-1})dx \quad [A] \\
 &= 0 - nr \int_0^1 [(1 - x^r)^n - 1](1 - x^r)^{n-1} dx \\
 &= 0 - nr \int_0^1 (1 - x^r)^n - (1 - x^r)^{n-1} dx \\
 I_n &= nr(-I_n + I_{n-1}) \\
 (nr + 1)I_n &= nrI_{n-1} \\
 I_n &= \frac{nr}{nr + 1} I_{n-1} \quad [B]
 \end{aligned}$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly equates I_1 , or equivalent merit.	1

Sample answer:

$$\begin{aligned}
 \text{For } r = \frac{3}{2} \text{ and } n = 3 \\
 I_3 &= \frac{3 \times \frac{3}{2}}{3 \times \frac{3}{2} + 1} \times I_2 \quad I_2 = \frac{2 \times \frac{3}{2}}{2 \times \frac{3}{2} + 1} \times I_1 \quad I_1 = \frac{1 \times \frac{3}{2}}{1 \times \frac{3}{2} + 1} \times I_0 \quad [A] \\
 \text{But } I_0 &= \int_0^1 (1 - x^r)^0 dx = \int_0^1 1 dx = 1 \\
 I_3 &= \frac{3 \times \frac{3}{2}}{3 \times \frac{3}{2} + 1} \times \frac{2 \times \frac{3}{2}}{2 \times \frac{3}{2} + 1} \times \frac{1 \times \frac{3}{2}}{1 \times \frac{3}{2} + 1} \times 1 \\
 &= \frac{81}{220} \quad [B]
 \end{aligned}$$

Question 16 (b) (i)

Criteria	Marks
Provides correct solution	2
Correctly uses the triangle inequality once to find either $x > \frac{5}{6}$, $x < \frac{3}{2}$ or $6 > -1$	1

Sample answer:

$$2x - 1 + 3x + 1 > 5 - x$$

$$6x > 5$$

$$x > \frac{5}{6} \quad (1) \quad \boxed{A} \text{ (or below)}$$

$$2x - 1 + 5 - x > 3x + 1$$

$$3 > 2x$$

$$x < \frac{3}{2} \quad (2) \quad \boxed{A} \text{ (or above)}$$

$$3x + 1 + 5 - x > 2x - 1$$

$$6 > -1 \quad \text{ie true regardless of the value of } x \quad \boxed{A} \text{ (or above)}$$

From (1) and (2):

$$\frac{5}{6} < x < \frac{3}{2} \quad \boxed{B}$$

Question 16 (b) (ii)

Criteria	Marks
Provides correct solution	2
Shows that $x = 1$ is the only answer that satisfies (i)	1

Sample answer:

If $2x - 1 = 3x + 1 \rightarrow x = -2$ which is outside the domain from (i) so $2x - 1 \neq 3x + 1$

If $2x - 1 = 5 - x \rightarrow x = 2$ which is outside the domain from (i) so $2x - 1 \neq 5 - x$

If $5 - x = 3x + 1 \rightarrow x = 1$ which is in the domain from (i) [A]

When $x = 1$, $5 - x = 3x + 1 = 4$ and $2x - 1 = 1$

$$\begin{aligned}\cos \theta &= \frac{1^2 + 4^2 - 4^2}{2 \times 1 \times 4} && [B] \\ &= \frac{1}{8}\end{aligned}$$

Alternative Solution:

If $2x - 1 = 3x + 1 \rightarrow x = -2$ which is outside the domain from (i) so $2x - 1 \neq 3x + 1$

If $2x - 1 = 5 - x \rightarrow x = 2$ which is outside the domain from (i) so $2x - 1 \neq 5 - x$

If $5 - x = 3x + 1 \rightarrow x = 1$ which is in the domain from (i) [A]

When $x = 1$, $5 - x = 3x + 1 = 4$ and $2x - 1 = 1$

Let D be the midpoint of AC . Since $\triangle ABC$ is isosceles with B as the apex, $BD \perp AC$.

$$\begin{aligned}\therefore \cos \theta &= \frac{AD}{AB} \\ &= \frac{\frac{1}{2} \times AC}{AB} \\ &= \frac{0.5}{4} && [B] \\ &= \frac{1}{8}\end{aligned}$$

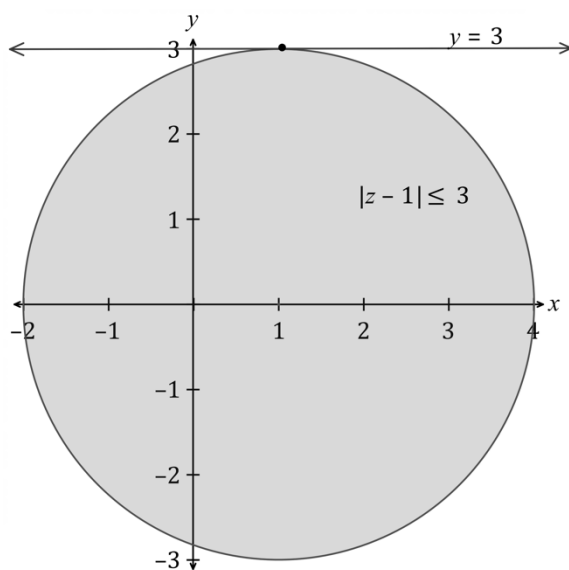
Question 16 (c)

Criteria	Marks
Provides correct solution	2
Correctly graphs both inequalities.	1

Sample answer:

$|z - 1| \leq 3$ represents a region with centre $(1, 0)$ and radius less than or equal to 3.
 $\text{Im}(z) \geq 3$ represents a region above the horizontal line $y = 3$.

A



The point $(1, 3)$ is where the two inequalities hold. **B**

Question 16 (d) (i)

Criteria	Marks
• Provides correct solution	4
• Shows that if $m^2 + n^2 + p^2 = r^2 + R^2$ then the circle C is on S_3 .	3
• Shows that if $m^2 + n^2 + p^2 = r^2 + R^2$ that $\triangle APB$ is right angled	2
• Finds $d^2 = m^2 + n^2 + p^2$, or equivalent merit	1

Sample answer:

$$d^2 = |\overrightarrow{AB}|^2 = (a + m - a)^2 + (b + n - b)^2 + (c - p - c)^2$$

$$= m^2 + n^2 + p^2 \quad (1) \quad \boxed{A}$$

Case 1: If $m^2 + n^2 + p^2 = r^2 + R^2$

From (1): $d^2 = r^2 + R^2 \quad (2)$

Let P be any point on C .

From (2): $|\overrightarrow{AB}|^2 = |\overrightarrow{AP}|^2 + |\overrightarrow{PB}|^2$ so $\triangle APB$ is right angled at $P \quad \boxed{B}$

$$\therefore \overrightarrow{AP} \cdot \overrightarrow{PB} = 0$$

$\therefore P$ lies on the sphere S_3 with diameter AB

Since P is any point on C then C lies on S_3 if $m^2 + n^2 + p^2 = r^2 + R^2. \quad \boxed{C}$

Case 2: If $m^2 + n^2 + p^2 \neq r^2 + R^2$

From (1): $d^2 \neq r^2 + R^2 \quad (3)$

$\therefore |\overrightarrow{AB}|^2 \neq |\overrightarrow{AP}|^2 + |\overrightarrow{PB}|^2$ so $\triangle APB$ is not right angled at P

$$\therefore \overrightarrow{AP} \cdot \overrightarrow{PB} \neq 0$$

$\therefore P$ does not lie on the sphere S_3 .

Since P is any point on C then C does not lie on S_3 if $m^2 + n^2 + p^2 \neq r^2 + R^2. \quad \boxed{D}$

$\therefore$ the circle C lies on the sphere S_3 whose diameter is AB

only if $m^2 + n^2 + p^2 = r^2 + R^2$

Question 16 (d) (ii)

Criteria	Marks
Provides correct solution	1

Sample answer:

$x^2 + y^2 + z^2 = 25$ is a sphere with centre (0,0,0) and radius 5, so $r = 5$.

$(x - 3)^2 + (y - 4)^2 + (z - 12)^2 = 144$ is a sphere with centre (3, 4, 12) and radius 12,
so $m = 3, n = 4, p = 12$ and $R = 12$.

$$m^2 + n^2 + p^2 = 3^2 + 4^2 + 12^2 = 169$$

$$r^2 + R^2 = 5^2 + 12^2 = 169$$

$$\therefore m^2 + n^2 + p^2 = r^2 + R^2 \quad \boxed{A}$$

So from (i) we can see that the two spheres intersect in a circle that lies on a sphere with a diameter whose ends are at the centres of the spheres.